

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (EC) Examination

November 2012

EC302 Integrated Circuits & Applications

Date: 27.11.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) In offset voltage compensating network design, why we take $R_b > R_{max} > R_c$. [03]

(b) Fig. (1) shows a differential amplifier using ideal op-amp. [04]

(a) Find the output voltage V_o .

(b) Show that the output corresponding to common-mode voltage

$$V_{cm} = \frac{(v_1 + v_2)}{2} \text{ is zero if } \frac{R'}{R} = \frac{R_2}{R_1}. \text{ Find } V_o \text{ in this case.}$$

Q - 2 (a) In the circuit of Fig. (2), $V_1=2$ V, $V_2=3$ V, $V_3=4$ V, $V_4=5$ V, $R_1=40$ K, $R_2=25$ K, $R_3=10$ K, $R_4=20$ K, $R_5=30$ K and $R_f=50$ K. Determine the output voltage V_o . [06]

(b) Write a short note on log Amplifier. [05]

(c) Describe the operation of current series negative feedback amplifier. [03]

OR

Q - 2 (a) What are the disadvantages of comparators? How it can be overcome? Explain in detail. [06]

(b) (i) Design an op-amp differentiator that will differentiate an input signal with $f_{max}=100$ Hz. [05]

(ii) Draw the output waveform for a sine wave of 1V peak at 100 Hz applied to the differentiator.

(iii) Repeat part (ii) for a square wave input.

(c) Why in peak detector, we always take $CR_d \leq \frac{T}{10}$ & $CR_L \geq 10T$. [03]

Q - 3 (a) Define Schmitt trigger circuit, explain its operation, and discuss its importance. [06]

- (b) Explain the Monolithic PLL & Discrete PLL in detail. [04]
 (c) Design a offset voltage compensation network for the LM741 op-amp. The op-amp uses ± 13 V supply voltage. [04]

OR

- Q - 3 (a) What is an absolute value output circuit? How can it be used as a full-wave rectifier? [06]
 (b) Derive the equation of output offset voltage in terms of input bias current. [05]
 (c) Define CMRR & explain the significance of a relatively large value of CMRR. [03]

SECTION - II

- Q - 4 (a) Explain $\alpha_{\max}$ and $\alpha_{\min}$ in terms of filter design. [02]
 (b) Define "order of filter". Explain its importance on performance of filter. [02]
 (c) Briefly explain the roles of a low pass filter and VCO in PLLs. [03]
 Q - 5 (a) Design the Biquad filter as directed: [07]
 Kind of filter: AP & BP for $\omega_0=10,000$ rad/sec, $Q=10$ & $C=0.1\mu\text{F}$.
 (b) What is a switching regulator? List four major components of the switching regulator. [04]
 (c) Using LM317, Design an adjustable voltage regulator to satisfy the following [03]
 specifications: output voltage $V_0=7.7$ V to 20.75 V, output current $I_0=0.5$ A,
 $I_{\text{ADJ}}=100\mu\text{A}$, $R_1=240\Omega$ and $V_{\text{REF}}=1.25$ V.

OR

- Q - 5 (a) Design a circuit to realize the transfer function of the Bode plot given in Fig. (3). Use inverting configuration. [07]
 (b) Draw Delyianis & friend circuit and determine its transfer function. Find ω_0, Q and BW in terms of R_1, R_2 and C . [07]
 Q - 6 (a) For the 555 timer as a astable multivibrator, $R_a=2.2\text{ k}\Omega$, $R_b=3.9\text{ k}\Omega$, and $C=0.1\mu\text{F}$. [06]
 Determine
 (i) The positive pulse width T_c ,
 (ii) Negative pulse width T_d , and
 (iii) Free-running frequency f_0 .
 (b) Draw the circuit for Biquad Band Stop Filter. Derive the transfer function and prepare its magnitude response and also draw the locations of poles & zeros on the s-plane. [08]

OR

- Q - 6 (a) List one application of the PLL and then briefly describe the role of the PLL in that application. [06]
- (b) Compare the Butterworth and Chebyshev responses. [04]
- (c) Explain application of 555 timer IC as 'a divide by 4' frequency divider. And mention clearly the relationship between the pulse width t_p and the period T of the input trigger signal. [04]

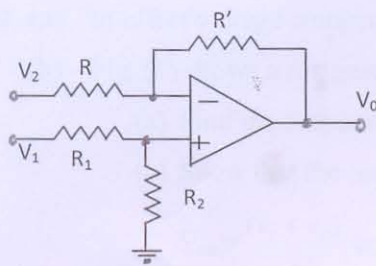


Fig. 1

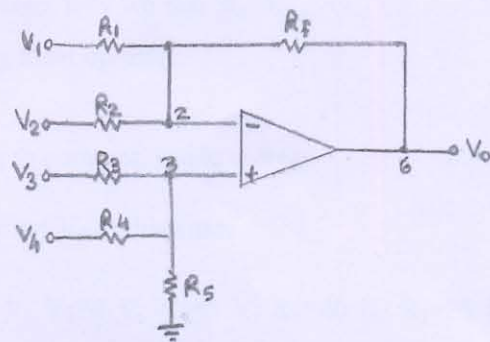


Fig. 2

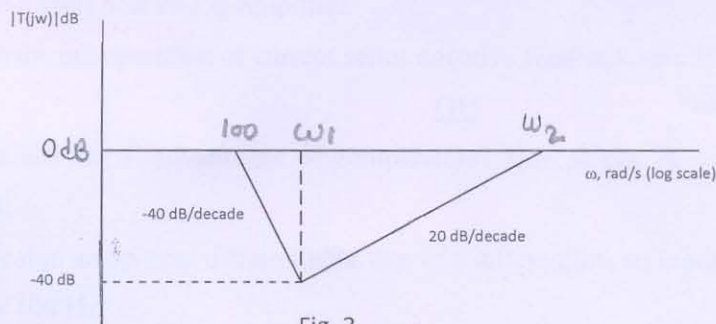


Fig. 3